

POLITECNICO
MILANO 1863

CASE STUDY - RED RIVER BASIN

Hoa Binh reservoir: identification of a forecast model of the inflows and optimal allocation of water resources

Riccardo Cuneo
Alessandro Prece
Andrea Romanò

Course of
NATURAL RESOURCES MANAGEMENT

Environmental and Land planning Engineering – A.Y. 2021-2022
Prof. A.Castelletti
Prof. M.Giuliani
Tutor M. Sangiorgio

CONTENTS

1. ABSTRACT

2. CONSTRUCTION OF A FORECAST MODEL FOR 5-DAY INFLOW

2.1 DATA VISUALIZATION AND EXPLORATION
2.2 MODELLING
2.2.1 LINEAR MODEL
2.2.2 ANN
2.2.3 CART
2.2.4 RF
2.3 CHOICE OF THE BEST MODEL

3. OPTIMAL ALLOCATION OF WATER RESOURCES

3.1 DEFINITION OF THE OBJECTIVES
3.2 CURRENT PERFORMANCE QUANTIFICATION
3.3 OPTIMIZATION VIA EMODPS

4. REFERENCES

1. ABSTRACT

The first part of this project aims at modelling the cumulative 5-day inflow to the Hoa Binh reservoir in the Red River basin in South-East Asia, between China and Vietnam.
The second part aims at optimally allocating water resource through an Evolutionary Multi-Objective Direct Policy Search Algorithm (EMODPS), accounting for contrasting interests between hydropower production at Hoa Binh and flood prevention in Hanoi.

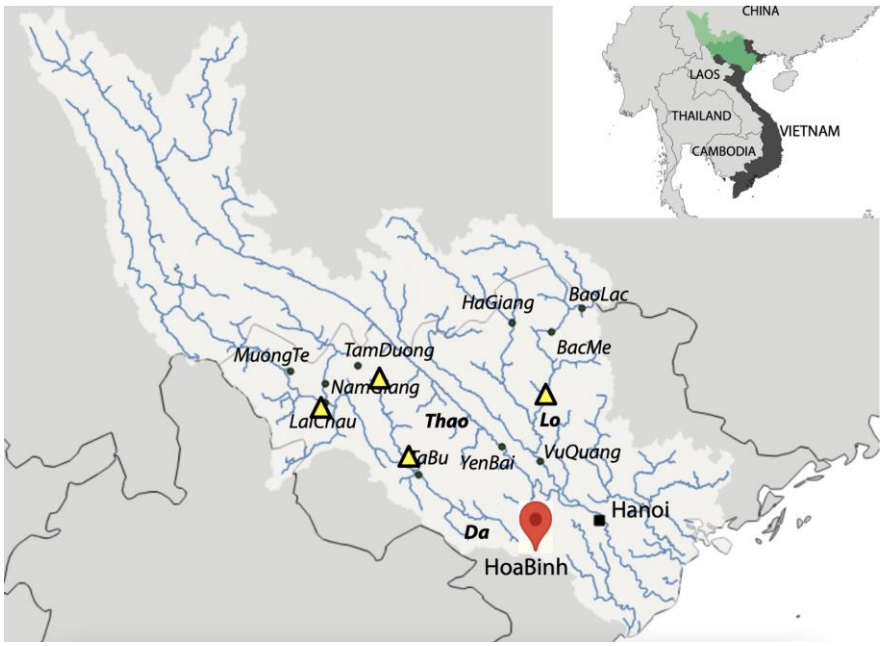


Fig. 1 - Red River Basin

2. CONSTRUCTION OF A FORECAST MODEL FOR 5-DAY INFLOW

2.1 DATA VISUALIZATION AND EXPLORATION

Two datasets are provided; a training one made of seven years and a test one of five years. Input variables are historical data of 7 years, as reported below. The output historical series is also given to construct the model.

DATA AVAILABLE (historical data)

Input variables

- Hoa Binh storage [m³]
- Hoa Binh inflow [m³/s]
- Flow at TaBu [m³/s]
- Flow at LaiChau [m³/s]
- Flow at NamGiang [m³/s]
- Flow at YenBai [m³/s]
- Flow at VaQuang [m³/s]
- Precipitation at MuongTe [m³/s]
- Precipitation at TamDuong [m³/s]
- Precipitation at Da [m³/s] (spatial average over the river basin)
- Precipitation at BaoLac [m³/s]
- Precipitation at BacMe [m³/s]
- Precipitation at HaGiang [m³/s]

Output variable

- Cumulative 5-day inflow at Hoa Binh [m³/s]

Firstly, it can be useful to visualise the input and output time series in order to observe their behaviour. Some of the variables are reported here.

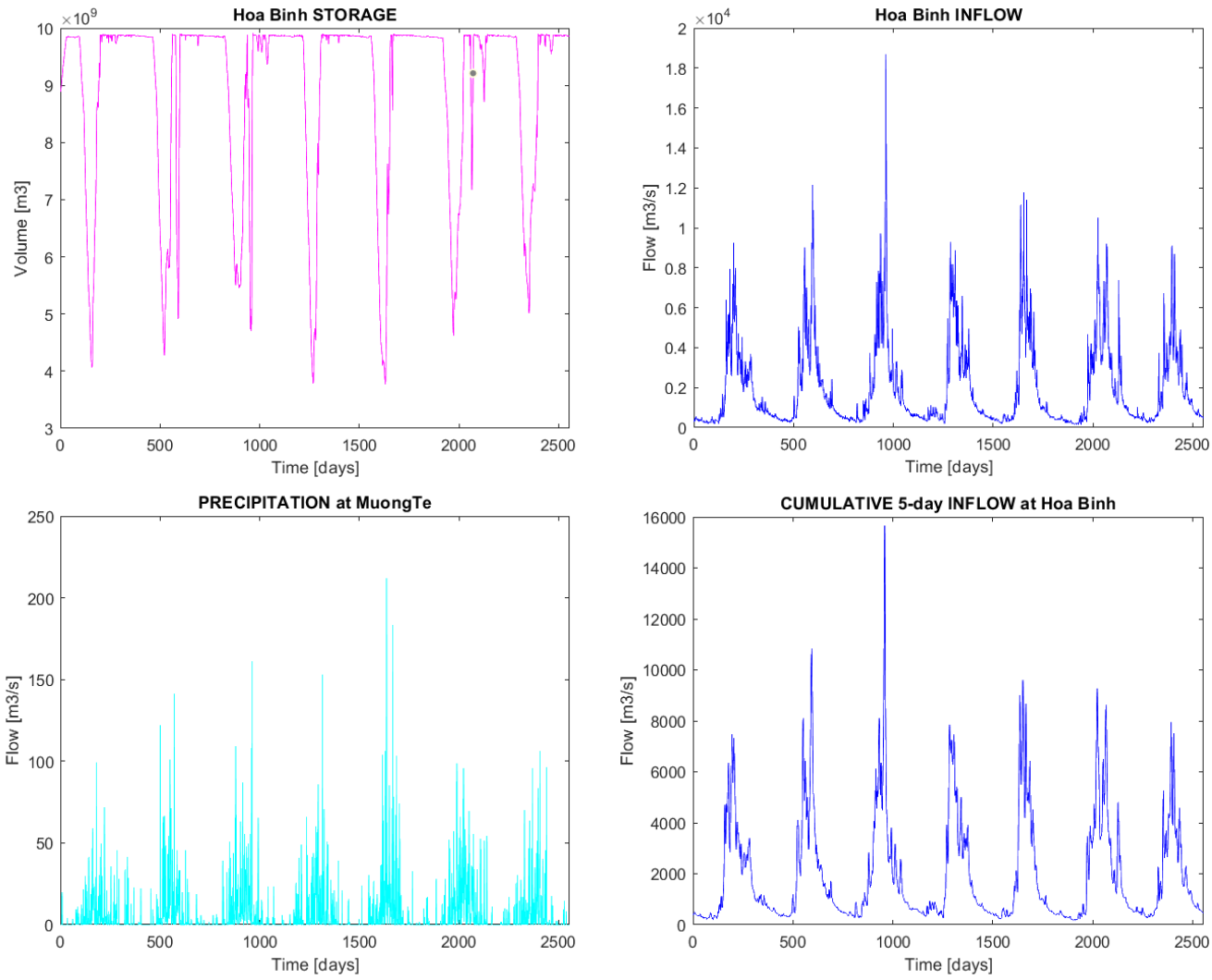


Fig. 2 – examples of some input trajectories, on the bottom right the output trajectory

Moreover, a correlogram has been used to assess the importance of the inputs at determining the output of the model.
What can be observed is that the flow inputs are the ones which have a significant correlation with the output. On the contrary, precipitation inputs show a non-correlation with the output.

Finally, storage time series has been removed from the dataset, as it is not a valid input for the model.

2.2 MODELLING

Looking at the data time series, one can observe a periodic behaviour, suggesting the presence of cyclo-stationarity in the data. This allows to remove the trend in the data according to the Box&Jenkins method. At the end of the modelling, it will be re-added to the data.

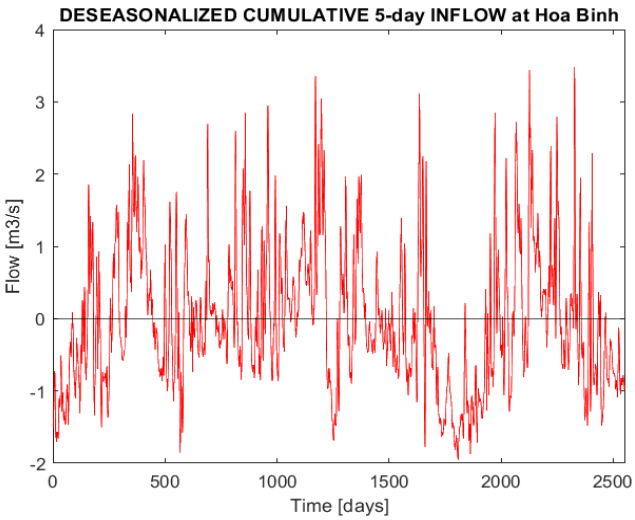


Fig. 3 – example of a deseasonalized output trajectory

The dataset is split into two parts: first five years for training and the last two years for validation. Now the modelling exercise can be performed, exploring different types of models (linear and non linear ones).

2.2.1 LINEAR MODEL

The first attempt was done by using a linear formulation with all the columns of the dataset and a Linear Least Squares optimization method for the parameters.

$$y = a_1x_1 + a_2x_2 + \ldots + a_nx_n$$

Then, considering the importance of the variables as shown in the correlogram, a linear model with less inputs, i.e. only the flows, has been tried.
A third attempt considered the precipitation inputs, too, and in particular with a delay of one time step. The results were not so satisfactory, inducing to think that a linear model was not suitable for this application.

2.2.2 ANN

A non-linear model should perform better on this type of environmental data, thus an Artificial Neural Network (ANN) can be a good solution. A feedforward net was used with the following parametrization:

number of hidden layers	1
neurons per hidden layer	5
train function	'trainlm'

Multiple calibrations were run to choose the best performing network (30 runs). The model performance was better than the linear case, reaching R^2 values of about 0.5 in validation.

2.2.3 CART

A Classification and Regression Tree (CART) model was used to try a non-parametric approach, searching for a comparable solution which required less computational cost than ANN. An automatic optimization of the hyper-parameters was applied. However, a CART model resulted to perform in a comparable way as the linear models ($R^2 < 0.4$).

2.2.4 RF

Another attempt was made through a Random Forest (RF) algorithm, applying a multiple-calibration procedure (30 runs). The parametrization of the RF algorithm was chosen as follow:

termination criterion	'MinLeafSize' = 10
ensemble aggregation method	'Bag' (bootstrap aggregation)
number of learning cycles	20

In this way, an R^2 score of about 0.5 was obtained in validation, that is comparable with the ANN performance.

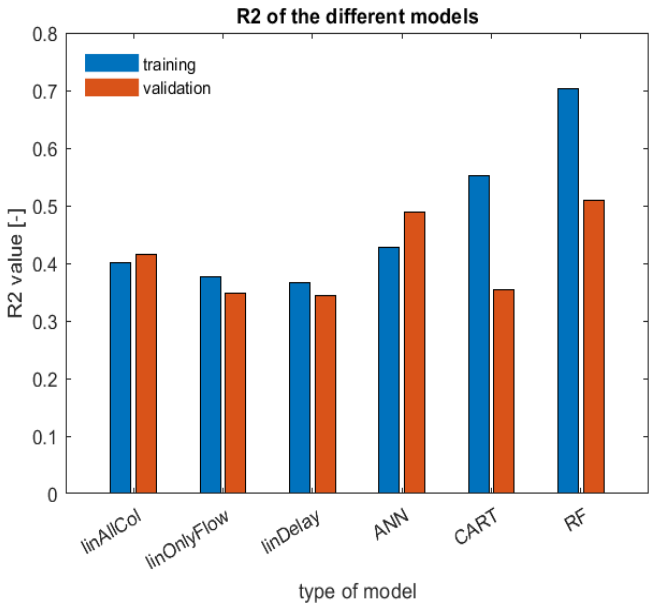


Fig. 4 – bar plot of R^2 using different models both in training and validation step

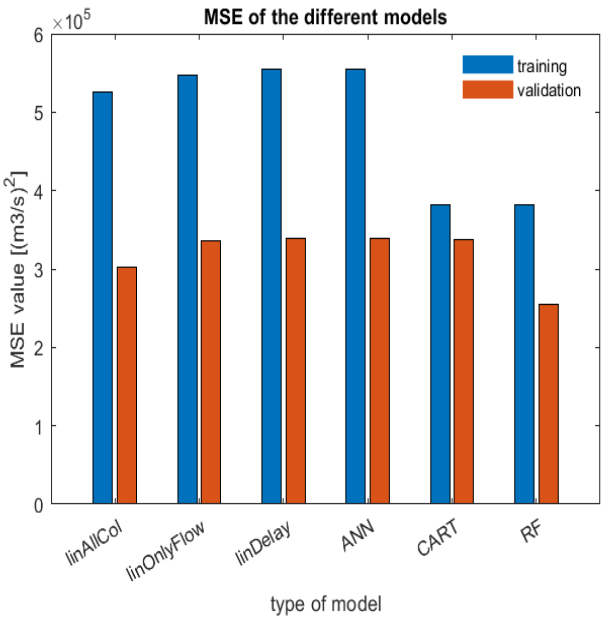


Fig. 5 – bar plot of MSE using different models both in training and validation step

MSE allows the parametrization to perform better on high values of output, rather than low values because it gives more weight to large errors than lowest one. According to this, as it is shown in the figure 5, MSE of the RF model reaches lower values compared to the others, which means that the distance J between prediction and observation has been minimised.

2.3 CHOICE OF THE MODEL

In order to choose the best model, the R^2 metric was considered as the first criterion to assess the model performance. The second criterion was the model complexity, in particular the computational cost. Considering R^2 score, only ANN and RF can be taken into account, as linear models and CART were not so performing, i.e. ANN and RF can describe better the non-linearities compared to the other models. As both the model took as input all the input dataset columns, and considering the 'not huge' amount of data, among these two models, ANN took more time to run the multiple calibrations, compared to RF, leading probably to a 'over-complexity' in the modelling exercise (time for run of the multiple-calibrated ANN 120.1 sec, running it using a standard laptop).

Thus, the RF model was chosen, being the best one at describing the output, taking into account its computational time too (time for run of the multiple-calibrated RF 10.3 sec seconds). Choosing another model would have only led to performance decrease and/or additional unnecessary computational costs.

Finally, MSE metric values strengthened the choice of RF model.

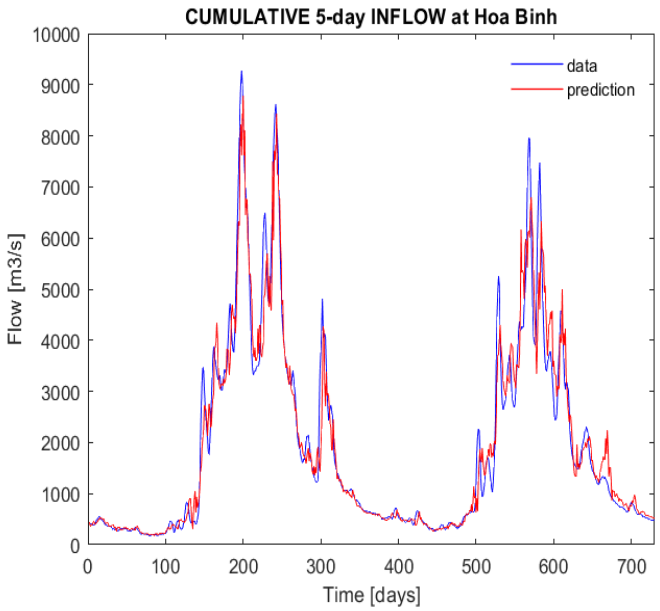


Fig. 6 – comparison between data and predicted trajectories in validation on the training dataset

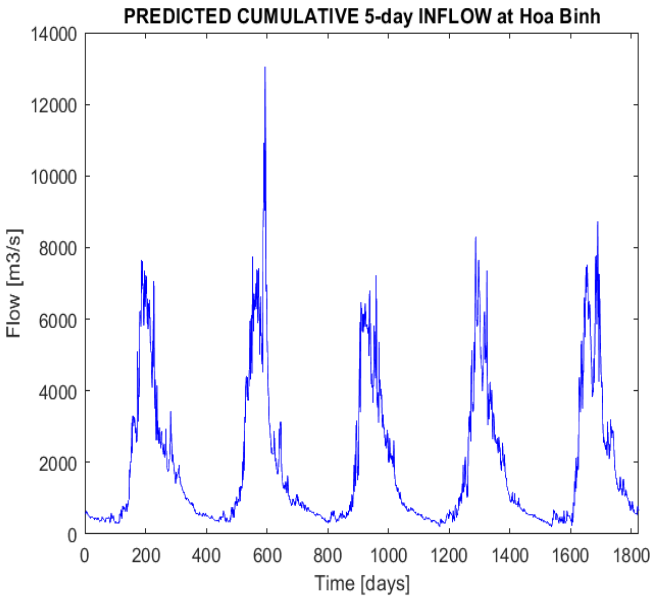


Fig. 7 – predicted trajectory of the output on the test dataset

3. OPTIMAL ALLOCATION OF WATER RESOURCES

The second part of the project aims at finding the best optimization for the operation of the multi-purpose Hoa Binh reservoir: choosing the best compromise solution between both hydropower production and flood protection in Hanoi.

Multi objective evolutionary algorithms were used to identify the optimal Standard Operating Policy (SOP) for the reservoir.

3.1 DEFINITION OF THE OBJECTIVES

The two objectives, one relative to hydropower production and the other relative to flood protection, are provided as follows:

- Hydropower production → mean daily production in [kWh/day] (***to be MAXIMIZED***, in graphs it has been considered with the opposite sign)

$$J^{HP} = \frac{1}{H}\sum_{t=1}^{H} P_t$$

- Flood control → mean daily squared water level excess in Hanoi in [cm^2] (***to be MINIMIZED***)

$$J^{flo} = \frac{1}{H}\sum_{t=1}^{H} F_t$$

3.2 CURRENT PERFORMANCE QUANTIFICATION

The first task is the quantification of the current policy performance, in order to be able to evaluate the improvement after the optimization procedure. The release from Hoa Binh reservoir is computed through the function '*std_operating_policy.m*' and its operation is simulated through the function '*simHB.m*'.

The simulated trajectories can be seen in *Fig. 8.*

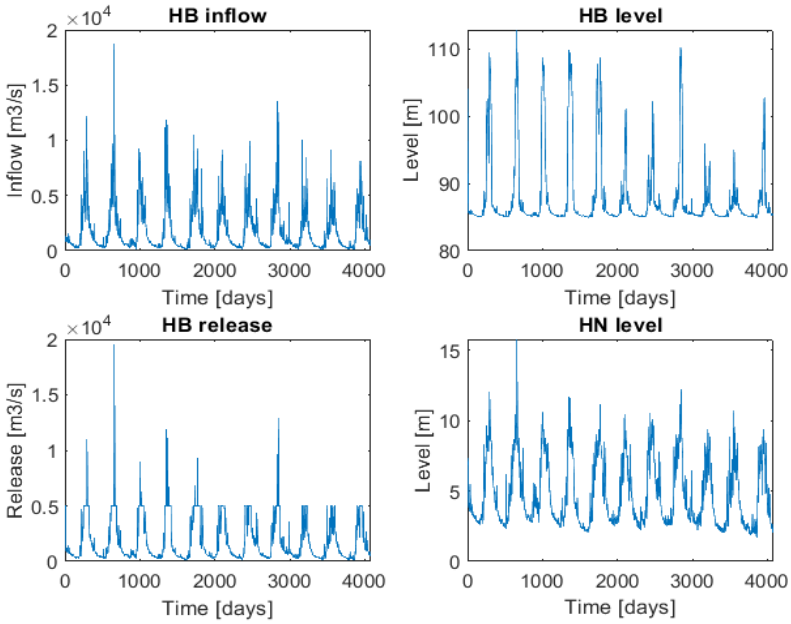


Fig. 8 – trajectories of Hoa Binh reservoir operations and water level in Hanoi

The current standard operating policy gives the following objective values:

J_HP	1.7 *10^7 [kWh/day]
J_flo	570 [cm^2]

Finally, the current standard operating policy can be represented.

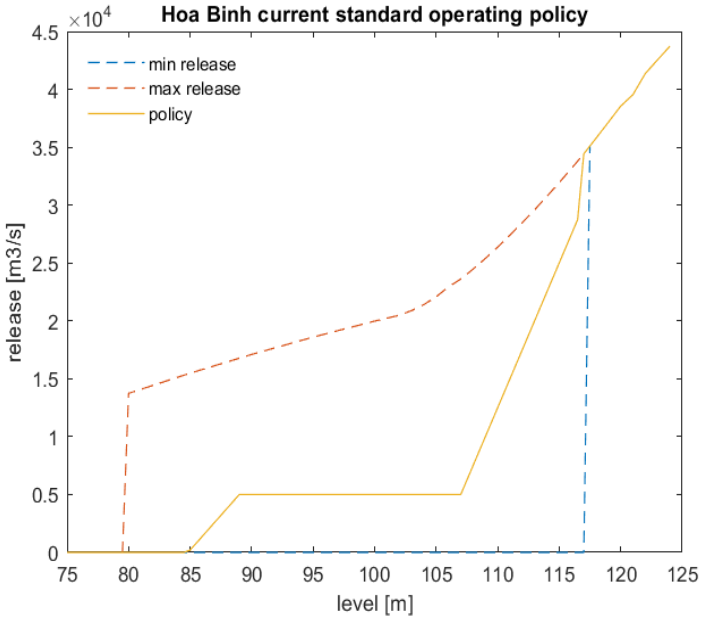


Fig. 9 – representation of the current standard operating policy

3.3 OPTIMIZATION VIA EMODPS

The optimization procedure has been performed through EMODPS method, using NSGAII algorithm several times. The results, i.e. the final populations, of the different runs of NSGAII are represented with different colours in Fig. 10. Then, a Pareto frontier was found out using the function '*find_pareto_frontier.m*'; the final results are reported in Fig. 11 where red dots are the values which are not dominated.

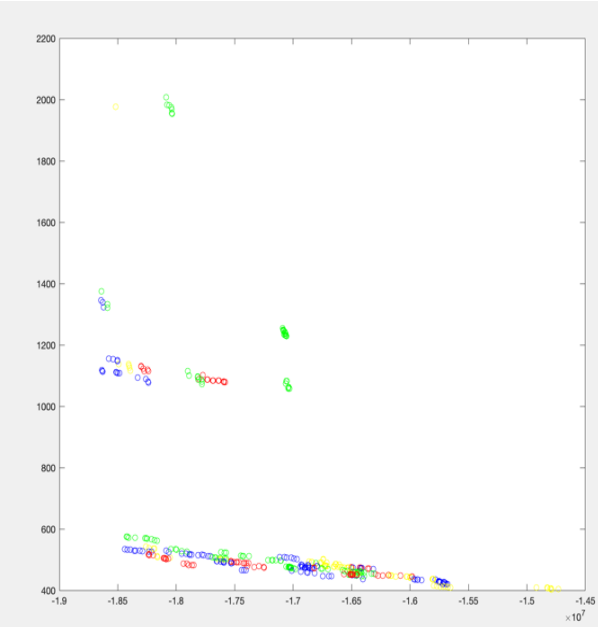


Fig. 10 – final populations of different runs of NSGAII (each color is a run)

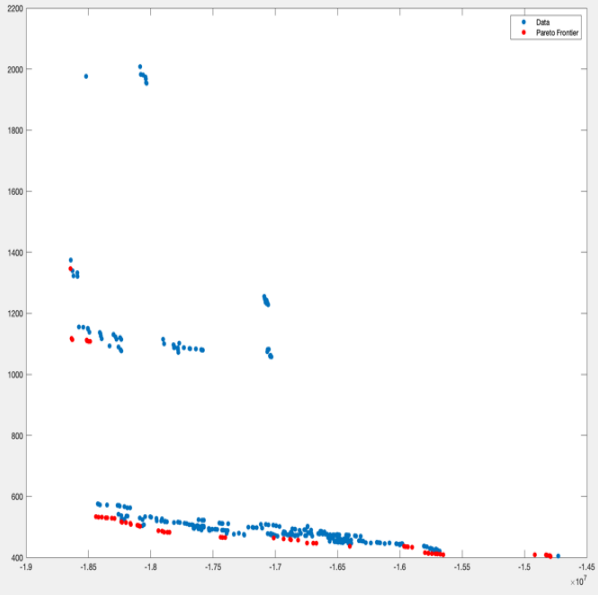


Fig. 11 – same points of Fig. 10 but in blue the dominated ones and in red the Pareto frontier

Blue dots are deleted, because dominated; the current policy is added to the graph. The more red dots are in the right hand part of the graph and the best is for floods, on the contrary, the more they are in the left hand the best is for the hydropower sector. Two alternatives are chosen, the first one gives the best performance to flood, the second one to hydropower.

Although these solutions give high performances to a single sector, they are bad for the other one. Thus, a compromise solution has to be found.

A compromise solution can be the one which increase at least the performance of a sector without decreasing the other one with respect to the current policy.

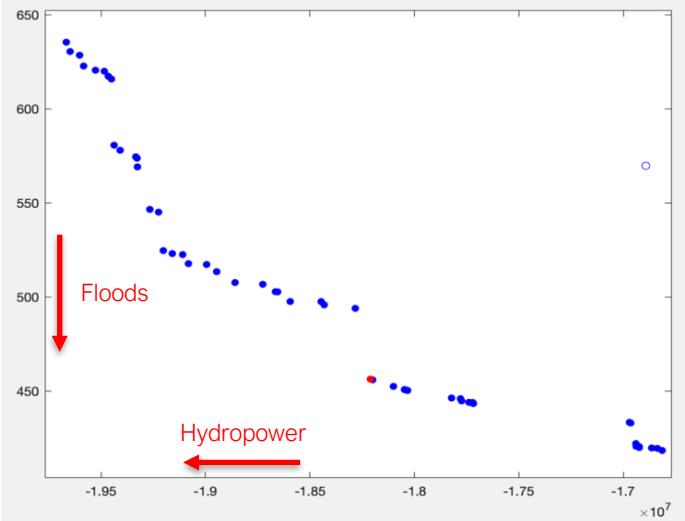


Fig. 12 – Pareto front in blue, the white dot is the current policy and the red one is the selected compromise solution

In the following table the objective values are reported for three policies, the best solution for hydropower, the compromise and the best for flooding one:

	J_HP [kWh/day]	J_flo [cm²]
Best hydropower	2.0 *10^7	4004
Compromise	1.8 *10^7	456
Best flooding	1.5 *10^7	382

In Fig. 13 the standard operating policy of the compromise solution can be seen.

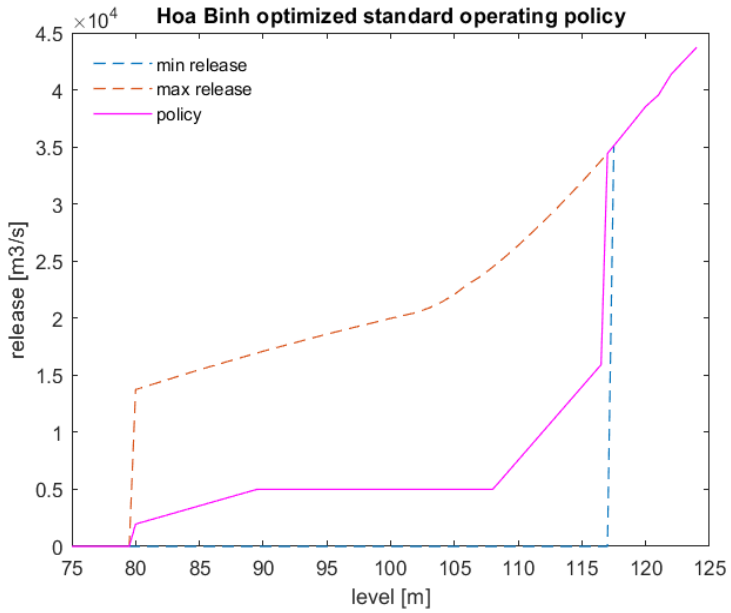


Fig. 13 – representation of the standard operating policy for the compromise solution

4. REFERENCES

- A. Castelletti, M. Giuliani, *Course lectures and notes*, 2022
- M. Sangiorgio, *Course Lab sessions*, 2022
- MathWorks documentation, *https://it.mathworks.com*